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$$f = \sum_{i=0}^n a_i x^i$$

The proof for Eisenstein p is prime $f \in \mathbb{Z}[x]$, $\partial f = n$ s.t. $\partial p \nmid a_1$

$\textcircled{1} p \mid a_i \text{ (s.t. } i < n) \text{ } \textcircled{2} p^2 \nmid a_0$ then f is reducible

assume $f = gh$, $g = \sum_{i=0}^s b_i x^i$, $h = \sum_{i=0}^t c_i x^i \in \mathbb{Z}[x]$

then $p \mid b_0 c_0 = a_0$, $p^2 \nmid b_0 c_0$

so assume $p \mid b_0$, $p \nmid c_0$

$$p \mid a_1 = b_1 c_0 + b_0 c_1 \Rightarrow p \mid b_1$$

$$p \mid a_2 = b_2 c_0 + b_1 c_1 + b_0 c_2 \Rightarrow p \mid b_2$$

$$p \mid a_s = \dots \Rightarrow p \mid b_s \quad (s < n)$$

but $p \nmid a_n = b_s c_t \Rightarrow p \nmid b_s$ contradict: f is irreducible over $\mathbb{Z}$

$\mathbb{Z}[x] \xleftarrow[\text{reducibility prime}]{\text{quotient}} \mathbb{Q}$

$(f(x), g(x)) = d(x)$ - the great common divisor gcd

$(a, b) = d \rightarrow \text{辗转相除法 (Euclidean algorithm)}$

II Finite dimensional Vector space

classical Examples $(F^n, +, \cdot)$ $(C^n$ and $R^n)$

Def Let $V \neq \emptyset$ if $(V, +)$ is an abelian group and $F \times V \xrightarrow{\cdot} V$ satisfies

$$\textcircled{1} (ab)v = a(bv)$$

$$\textcircled{2} (a+b)v = av + bv$$

$$\textcircled{3} a(v+w) = av + aw$$

$$\textcircled{4} 1 \cdot v = v$$

RK1 "+" and "·" over F

RK2 $1 \cdot v = v$ (given)

Then $(V, +, \cdot)$ is called to be a vector space over F .

Examples. $\textcircled{1} F^n / F$ $\textcircled{2} F^{m \times n} / F$ $\textcircled{3} F[x], F[x]_n / F$

$\textcircled{4} C[a, b], R^n / R$ $\textcircled{5} L, R / R$ - dimension n

$$\textcircled{6} F^A = (\{f: A \rightarrow F\}, +, \cdot)$$

$$f+g: A \rightarrow F$$

$$a \mapsto (f+g)(a) = f(a) + g(a)$$

$$0: A \mapsto 0$$

$$-f: A \mapsto (-f)(a) = -f(a)$$

so $(F, +)$ is an abelian group

$$\lambda f: A \mapsto (\lambda f)(a) = \lambda f(a)$$

so $(F^A, +, \cdot)$ is a VS $|F$

$\{A \xrightarrow{f} B\} = B^A$ Let $|A|=m, |B|=n$ Then $|B^A| = n^m = |B|^{|A|}$

eg. $|A|=2$ $A=\{a_1, a_2\}$, $\dim F^A = 2 = |A|$

Analysis: $\forall f \in F^A, f = \sum_{i=1}^n x_i f_i$

$$f(a_j) = \delta_{ij} \quad f(a_j) = \sum_{i=1}^n x_i f_i(a_j) \quad j=1,2 \quad \begin{cases} f_i(a_1)=1, f_i(a_2)=0 \\ f_i(a_1)=0, f_i(a_2)=1 \end{cases}$$

Linear Independent (LI): $f(a_j) = \sum_{i=1}^n x_i f_i(a_j)$

$$\sum_{i=1}^n x_i f_i = 0 \Rightarrow x_i = 0, \text{ all } i$$

$$0 = \sum_{i=1}^n x_i f_i(a_j) = x_j f_j(a_j) = x_j$$

$$\text{故 } x_i = f_i(a_i)$$

In general, $A = \{a_1, \dots, a_n\}$. Find a basis of $\mathbb{R}^n$:

$$f_i(a_j) = \delta_{ij} \quad 1 \leq i \leq n$$

Def Let V be a vector space over $\mathbb{F}$. if there is $B = \{v_1, \dots, v_n\} \subseteq V$ s.t. B is LI and B is maximal (i.e. $\forall v \in V, \exists x_i \in \mathbb{F}$ s.t. $v = \sum_{i=1}^n x_i v_i$) Then B is a basis of V (otherwise, V is infinite dimensional)

Thm Let V be a vector space over $\mathbb{F}$ and $B = \{v_1, \dots, v_n\}$ a basis of V . Then any basis of V contains exactly n vectors, so this n is independent of the choice of bases and is called to be the dimension of V .

Pf. Let $C = \{u_1, \dots, u_m\}$ be another basis of V .

$$\text{then } u_j = a_{1j} v_1 + a_{2j} v_2 + \dots + a_{nj} v_n \quad (1 \leq j \leq m)$$

$$\text{Denote this by } \begin{matrix} u_j = (v_1, v_2, \dots, v_n) \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{pmatrix} \\ v_k = (u_1, u_2, \dots, u_m) \begin{pmatrix} b_{1k} \\ b_{2k} \\ \vdots \\ b_{mk} \end{pmatrix} \end{matrix} \quad \begin{matrix} A = (a_{ij})_{n \times m} \\ B = (b_{ik})_{m \times n} \end{matrix}$$

$$(u_1, u_2, \dots, u_m) = (v_1, v_2, \dots, v_n) B A \Rightarrow BA = I_n \quad AB = I_m \Rightarrow m = n$$